

Module 02: Agents in Embodied Cognition — The Intuitive Knower

Learning Objectives

1. Define the **embodied agent as an intuitive knower** — an inference system whose generative model is built from years of somatic experience rather than explicit propositional knowledge.
2. Analyze how **expertise, felt sense, and gut knowing** emerge from deep generative models with high-precision priors shaped by practice.
3. Apply the Active Inference framework to understand expert intuition across diverse domains (clinical, athletic, creative, emergency).

Introduction

"I just *knew* something was wrong." Every experienced practitioner — surgeon, firefighter, midwife, martial artist, therapist — describes moments of intuitive knowledge that bypass conscious deliberation. This is not mysticism; it is the product of a **deeply refined generative model**. Years of practice have tuned the agent's priors with such precision that inference happens faster than verbal reasoning can track. The prediction error that signals danger arrives as a gut feeling, not a conclusion.

This module examines the embodied agent as an intuitive knower — how the body-mind system generates high-confidence inferences through somatic channels that precede and sometimes override cognitive analysis.

Key Concepts

1. Expertise as Generative Model Depth

Novices rely on rules; experts rely on pattern recognition. In Active Inference terms, the expert's generative model has been trained on thousands of episodes, producing:

- **Highly structured priors:** The expert's B matrix (transition model) encodes rich temporal regularities — "when I see this, the next thing that happens is usually that"
- **High-precision priors:** The expert's predictions are confident and specific, not vague — enabling fast inference that "skips" the deliberate processing required by novices
- **Deep hierarchical models:** The expert represents the situation at multiple levels simultaneously — tactical details, strategic patterns, and meta-cognitive awareness of their own uncertainty

Dreyfus's five-stage model of skill acquisition (Novice → Advanced Beginner → Competent → Proficient → Expert) maps directly onto the progressive deepening and precision-tuning of the generative model.

2. Felt Sense and Interoceptive Inference

Eugene Gendlin's concept of **felt sense** describes a bodily-experienced knowing that contains implicit information about a situation. In Active Inference terms, the felt sense arises when:

- Interoceptive prediction errors (visceral, autonomic, proprioceptive) accumulate into a coherent pattern
- The body's generative model has detected a discrepancy between the current situation and what is "safe" or "right," but the discrepancy has not yet been articulated verbally
- The felt sense is a **pre-verbal prediction error summary** — the body's compressed communication to the conscious mind that "something needs attention"

A veteran emergency room nurse who "feels" that a patient is about to deteriorate is processing dozens of micro-observations (skin color, breathing pattern, pupil response, smell) through a deeply trained interoceptive model. The conclusion arrives as a feeling before it can be expressed as a diagnosis.

3. Gut Decisions and Somatic Markers

Damasio's somatic marker hypothesis proposes that emotional feelings guide decision-making by providing fast interoceptive evaluations of options. In Active Inference terms:

- Each decision option triggers a **predicted interoceptive state** (the body simulates how it would feel to take each action)
- The option that minimizes predicted interoceptive free energy (feels most "right") is preferentially selected
- This explains why damage to the ventromedial prefrontal cortex (which integrates interoceptive signals) impairs decision-making even when intellectual reasoning is preserved

4. When Intuition Fails

Expert intuition is not infallible. It fails when:

- **The environment changes:** The expert's generative model was trained under conditions that no longer hold (a combat pilot's desert warfare intuitions fail in urban combat)
- **Prediction errors are systematically ignored:** High-precision priors can override genuine novel signals — the expert "sees what they expect" rather than what is there
- **Stress degrades interoceptive precision:** Under extreme stress, the signal-to-noise ratio of somatic markers drops, producing unreliable intuitions

Applications

- **Expert firefighter intuition:** A seasoned firefighter entering a burning building feels a gut-level "pull" to evacuate seconds before the floor collapses. The firefighter's generative model has accumulated deep priors from years of structural fire experience — heat patterns, sound changes, floor flexion — that generate a prediction of catastrophic surprise, enacted as immediate withdrawal before conscious reasoning articulates why.
- **Clinical palpation expertise:** A veteran midwife places her hands on a laboring mother's abdomen and intuitively knows the baby has shifted position, even though the palpation findings are ambiguous to a less experienced practitioner. Her Markov blanket

(hands, proprioceptive sensitivity, visceral resonance) is exquisitely tuned by decades of tactile inference.

Conclusion

The intuitive knower is not a mystic — they are an embodied Active Inference agent with a deeply refined generative model and high-precision interoceptive inference. Expertise, felt sense, and gut decisions are all manifestations of sophisticated somatic prediction. The next module examines how these intuitive agents perceive the world through embodied channels.